

Vector Calculus

Vector function

If to each value of a scalar variable t , there corresponds a value of a vector $\vec{r}$, then $\vec{r}$ is called a vector function of the scalar variable t and write $\vec{r} = \vec{r}(t)$ or, $\vec{r} = \vec{f}(t)$
eg. The position vector $\vec{r}$ of a particle moving along a curved path is a vector fn of time t .

Derivative of a vector function. w.r.t. a scalar.

Let $\vec{r} = \vec{f}(t)$ be a vector fn of the scalar variable t .
Let δt be a small increment in t and $\delta \vec{r}$, the corresponding increment in $\vec{r}$, then

$$\begin{aligned}\vec{r} + \delta \vec{r} &= \vec{f}(t + \delta t) \\ \Rightarrow \delta \vec{r} &= \vec{f}(t + \delta t) - \vec{r} \\ \Rightarrow \delta \vec{r} &= \vec{f}(t + \delta t) - \vec{f}(t) \\ \text{and } \frac{\delta \vec{r}}{\delta t} &= \frac{\vec{f}(t + \delta t) - \vec{f}(t)}{\delta t}\end{aligned}$$

If $\lim_{\delta t \rightarrow 0} \frac{\delta \vec{r}}{\delta t} = \lim_{\delta t \rightarrow 0} \frac{\vec{f}(t + \delta t) - \vec{f}(t)}{\delta t}$ exist, then the value of this limit is denoted by $\frac{d\vec{r}}{dt}$ and is called derivative of $\vec{r}$ w.r.t. t .

$\frac{d^2 \vec{r}}{dt^2}$ is called the second derivative of $\vec{r}$ w.r.t. t .

General rules for vector differentiation

Let $\vec{u}, \vec{v}, \vec{w}$ be vector function and ϕ be scalar fn of t , then

1. $\frac{d}{dt} (\alpha \vec{v}) = \alpha \frac{d\vec{v}}{dt}$, α being a constant
2. $\frac{d}{dt} (\vec{u} + \vec{v}) = \frac{d\vec{u}}{dt} + \frac{d\vec{v}}{dt}$
3. $\frac{d}{dt} (\phi \vec{v}) = \frac{d\phi}{dt} \vec{v} + \phi \frac{d\vec{v}}{dt}$
4. $\frac{d}{dt} (\vec{u} \cdot \vec{v}) = \frac{d\vec{u}}{dt} \cdot \vec{v} + \vec{u} \cdot \frac{d\vec{v}}{dt}$ (dot product)
5. $\frac{d}{dt} (\vec{u} \times \vec{v}) = \frac{d\vec{u}}{dt} \times \vec{v} + \vec{u} \times \frac{d\vec{v}}{dt}$ (Cross product)
6. $\frac{d}{dt} (\vec{u} \cdot \vec{v} \cdot \vec{w}) = \left(\frac{d\vec{u}}{dt} \cdot \vec{v} \cdot \vec{w} \right) + \left(\vec{u} \cdot \frac{d\vec{v}}{dt} \cdot \vec{w} \right) + \left(\vec{u} \cdot \vec{v} \cdot \frac{d\vec{w}}{dt} \right)$
7. $\frac{d}{dt} (\vec{u} \times \vec{v} \times \vec{w}) = \frac{d\vec{u}}{dt} \times (\vec{v} \times \vec{w}) + \vec{u} \times \left(\frac{d\vec{v}}{dt} \times \vec{w} \right) + \vec{u} \times (\vec{v} \times \frac{d\vec{w}}{dt})$

Derivative of a Constant function

A vector is said to be constant if both its magnitude and direction are fixed.

- * If $\vec{F}(t)$ has a constant magnitude, then $\boxed{\vec{F} \cdot \frac{d\vec{F}}{dt} = 0}$
- * If $\vec{F}(t)$ has a constant direction, then $\boxed{\vec{F} \times \frac{d\vec{F}}{dt} = 0}$

Velocity and Acceleration

If the scalar variable t denotes the time and $\vec{r}$ is the position vector of a moving particle P , then $\delta\vec{r}$ is the displacement of the particle in time δt .

The vector $\frac{\delta\vec{r}}{\delta t}$ is the average velocity of the particle during the interval δt . If $\vec{v}$ represent the velocity vector of the particle at P , then $\vec{v} = \lim_{\delta t \rightarrow 0} \frac{\delta\vec{r}}{\delta t} = \frac{d\vec{r}}{dt}$.

If $\vec{a}$ represents the acceleration of the particle at P , then $\vec{a} = \lim_{\delta t \rightarrow 0} \frac{\delta\vec{v}}{\delta t} = \frac{d\vec{v}}{dt} = \frac{d}{dt} \left(\frac{d\vec{r}}{dt} \right) = \frac{d^2\vec{r}}{dt^2}$.

Q. Show that if $\vec{r} = \vec{a} \sin \omega t + \vec{b} \cos \omega t$, where $\vec{a}, \vec{b}, \omega$ are constants then show that (a) $\frac{d^2\vec{r}}{dt^2} = -\omega^2 \vec{r}$ (b) $\vec{r} \times \frac{d\vec{r}}{dt} = -\omega \vec{a} \times \vec{b}$

Solⁿ. (a) $\vec{r} = \vec{a} \sin \omega t + \vec{b} \cos \omega t$

$$\therefore \frac{d\vec{r}}{dt} = \vec{a} \omega \cos \omega t - \vec{b} \omega \sin \omega t$$

$$\frac{d^2\vec{r}}{dt^2} = -\vec{a} \omega^2 \sin \omega t - \vec{b} \omega^2 \cos \omega t$$

$$= -\omega^2 (\vec{a} \sin \omega t + \vec{b} \cos \omega t)$$

$$= -\omega^2 \vec{r}$$

$$\begin{aligned} \text{(b)} \quad \vec{r} \times \frac{d\vec{r}}{dt} &= (\vec{a} \sin \omega t + \vec{b} \cos \omega t) \times (\vec{a} \omega \cos \omega t - \vec{b} \omega \sin \omega t) \\ &= [-\vec{a} \times \vec{b} \sin^2 \omega t + \vec{b} \times \vec{a} \cos^2 \omega t] \omega \quad \left[\begin{array}{l} \because \vec{a} \times \vec{a} = 0 \\ \vec{b} \times \vec{b} = 0 \end{array} \right] \\ &= (-\vec{a} \times \vec{b} \sin^2 \omega t - \vec{a} \times \vec{b} \cos^2 \omega t) \omega \\ &= -\omega \vec{a} \times \vec{b} \end{aligned}$$

Q. If $\frac{d\vec{u}}{dt} = \vec{\omega} \times \vec{u}$ and $\frac{d\vec{v}}{dt} = \vec{\omega} \times \vec{v}$, prove that

$$\frac{d}{dt} (\vec{u} \times \vec{v}) = \vec{\omega} \times (\vec{u} \times \vec{v})$$

Solⁿ. $\frac{d}{dt} (\vec{u} \times \vec{v}) = \vec{u} \times \frac{d\vec{v}}{dt} + \frac{d\vec{u}}{dt} \times \vec{v}$

$$= \vec{u} \times (\vec{\omega} \times \vec{v}) + (\vec{\omega} \times \vec{u}) \times \vec{v}$$

$$= (\vec{u} \cdot \vec{v})\vec{\omega} - (\vec{u} \cdot \vec{\omega})\vec{v} + (\vec{v} \cdot \vec{\omega})\vec{u} - (\vec{v} \cdot \vec{u})\vec{\omega}$$

$$= (\vec{u} \cdot \vec{v})\vec{\omega} - (\vec{u} \cdot \vec{\omega})\vec{v} + (\vec{v} \cdot \vec{\omega})\vec{u} - (\vec{u} \cdot \vec{v})\vec{\omega}$$

$$\boxed{\because \vec{u} \cdot \vec{v} = \vec{v} \cdot \vec{u}}$$

$$= (\vec{v} \cdot \vec{\omega})\vec{u} - (\vec{u} \cdot \vec{\omega})\vec{v}$$

$$= (\vec{\omega} \cdot \vec{v})\vec{u} - (\vec{\omega} \cdot \vec{u})\vec{v}$$

$$= \vec{\omega} \times (\vec{u} \times \vec{v})$$

$$\vec{A} \cdot \vec{B} = ab \cos \theta$$

$$\vec{A} \times \vec{B} = ab \sin \theta \hat{n}$$

$$\vec{a} \times (\vec{b} \times \vec{c})$$

$$= (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c}$$

and $(\vec{a} \times \vec{b}) \times \vec{c} =$

$$(\vec{c} \cdot \vec{a})\vec{b} - (\vec{c} \cdot \vec{b})\vec{a}$$

Q. A particle moves along a curve whose parametric eqn in terms of time t , is $x = 3t^2$, $y = t^2 - 2t$, $z = t^3$.

Find its velocity and acceleration at time $t = 2$

Solⁿ. If $\vec{r}$ is the position vector of the particle at any time t , then $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k} = 3t^2\vec{i} + (t^2 - 2t)\vec{j} + t^3\vec{k}$

Then the velocity $\vec{v}$ of the particle at a time t , is given by

$$\vec{v} = \frac{d\vec{r}}{dt} = 6t\vec{i} + (2t - 2)\vec{j} + 3t^2\vec{k}$$

and the acceleration $\vec{a}$ of the particle at time t is given by

$$\vec{a} = \frac{d^2\vec{r}}{dt^2} = 6\vec{i} + 2\vec{j} + 6t\vec{k}$$

At time $t = 2$, we have $\vec{v} = 12\vec{i} + 2\vec{j} + 12\vec{k}$

and $\vec{a} = 6\vec{i} + 2\vec{j} + 12\vec{k}$

Q. Find the angle between the tangents to the curve $\mathbf{r} = t^2\mathbf{i} + 2t\mathbf{j} + t^3\mathbf{k}$ at the points $t=1$ and $t=2$.

Solⁿ. $\rightarrow$ The tangent at any point 't' is given by

$$\frac{d\mathbf{r}}{dt} = 2t\mathbf{i} + 2\mathbf{j} + 3t^2\mathbf{k}$$

$\therefore$ The tangents $\vec{T}_1$ and $\vec{T}_2$ at $t=1$ and $t=2$, are given by

$$\vec{T}_1 = 2\mathbf{i} + 2\mathbf{j} + 3\mathbf{k}, \quad \vec{T}_2 = 4\mathbf{i} + 4\mathbf{j} + 12\mathbf{k}$$

If θ is the angle between $\vec{T}_1$ and $\vec{T}_2$, then

$$\cos\theta = \frac{\vec{T}_1 \cdot \vec{T}_2}{|\vec{T}_1| |\vec{T}_2|} = \frac{2(4) + (2)(4) + 3(12)}{\sqrt{2^2 + 2^2 + 3^2} \sqrt{4^2 + 4^2 + 12^2}} = \frac{48}{\sqrt{17} \sqrt{164}}$$

$$\therefore \theta = \cos^{-1}\left(\frac{24}{\sqrt{17} \sqrt{41}}\right) \text{ is the required angle.}$$

$$\left. \begin{array}{l} \mathbf{i} \cdot \mathbf{i} = 1 \\ \mathbf{j} \cdot \mathbf{j} = 1 \\ \mathbf{k} \cdot \mathbf{k} = 1 \\ \mathbf{i} \cdot \mathbf{j} = 0 \\ \mathbf{j} \cdot \mathbf{k} = 0 \\ \mathbf{k} \cdot \mathbf{i} = 0 \end{array} \right\}$$

Scalar and Vector Point Function

Point Function :- A variable quantity which depends for its value on its position only, i.e., upon the coordinates of the points, say (x, y, z) in space is called a point function.

There are two types of point functions.

1. Scalar Point Function :-

If a scalar quantity ϕ is a fn of a point, i.e., depends on its position (x, y, z) in space, then $\phi(x, y, z)$ is called a scalar point function. The scalar point fn does not depend upon the choice of coordinate system.

Ex - density of a body, the temperature of a body at any instant, potential of a body.

2. Vector Point Function :

If a vector quantity $\vec{V}$ depends for its value on its position (x, y, z) in space, then $\vec{V}(x, y, z)$ is called a vector point function.

Ex - Velocity of a moving fluid, force, gravitational force or electrical intensity

Gradient of a Scalar Point Function

The vector differential operator $\hat{i}\frac{\partial}{\partial x} + \hat{j}\frac{\partial}{\partial y} + \hat{k}\frac{\partial}{\partial z} = \nabla$

is denoted by the symbol ∇ and is called Del (or Nabla) when it operates on any scalar point function $\phi(x, y, z)$

say, we get a vector quantity $\vec{\nabla}\phi = \hat{i}\frac{\partial\phi}{\partial x} + \hat{j}\frac{\partial\phi}{\partial y} + \hat{k}\frac{\partial\phi}{\partial z}$

which is called the gradient of the scalar point function $\phi(x, y, z)$. This is also written as gradient ϕ or $\text{grad } \phi$

Thus $\boxed{\text{grad } \phi = \vec{\nabla}\phi = \hat{i}\frac{\partial\phi}{\partial x} + \hat{j}\frac{\partial\phi}{\partial y} + \hat{k}\frac{\partial\phi}{\partial z}}$

Properties of Gradient

- (a) If ϕ is a constant scalar point function, then $\nabla\phi = 0$
- (b) If ϕ_1 and ϕ_2 are two scalar point functions, then
 - (i) $\nabla(\phi_1 \pm \phi_2) = \nabla\phi_1 \pm \nabla\phi_2$
 - (ii) $\nabla(c_1\phi_1 \pm c_2\phi_2) = c_1\nabla\phi_1 \pm c_2\nabla\phi_2$, c_1, c_2 are const.
 - (iii) $\nabla(\phi_1\phi_2) = \phi_1\nabla\phi_2 + \phi_2\nabla\phi_1$
 - (iv) $\nabla(\phi_1/\phi_2) = \frac{\phi_2\nabla\phi_1 - \phi_1\nabla\phi_2}{\phi_2^2}$, $\phi_2 \neq 0$

Q. Find gradient ϕ when ϕ is given by

$$\phi = 3x^2y - y^3z^2 \text{ at the point } (1, -2, -1)$$

Solⁿ $\rightarrow$ $\text{Grad } \phi = \vec{\nabla} \phi = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \cdot (3x^2y - y^3z^2)$

$$= \hat{i} \frac{\partial}{\partial x} (3x^2y - y^3z^2) + \hat{j} \frac{\partial}{\partial y} (3x^2y - y^3z^2) + \hat{k} \frac{\partial}{\partial z} (3x^2y - y^3z^2)$$

$$= \hat{i} (6xy) + \hat{j} (3x^2 - 3y^2z^2) + \hat{k} (-2y^3z)$$

At $(x, y, z) \rightarrow (1, -2, -1)$ we get

$$\text{Grad } \phi = -12\hat{i} - 9\hat{j} - 16\hat{k}$$

Q. Find a unit ^{vector} normal to the surface $x^3 + y^3 + z^3 + 3xyz = 3$ at the point $(1, 2, -1)$

Solⁿ $\rightarrow$ Equation of the given surface is $\phi = x^3 + y^3 + z^3 + 3xyz = 3$

Since $\vec{\nabla} \phi$ is a vector normal to the surface, we have

$$\vec{\nabla} \phi = \hat{i} \frac{\partial \phi}{\partial x} + \hat{j} \frac{\partial \phi}{\partial y} + \hat{k} \frac{\partial \phi}{\partial z} = \hat{i} (3x^2 + 3yz) + \hat{j} (3y^2 + 3xz) + \hat{k} (3xy + 3z^2)$$

$$\therefore [\vec{\nabla} \phi]_{(1, 2, -1)} = \hat{i} (3 - 6) + \hat{j} (12 - 3) + \hat{k} (3 + 6) = 3(-\hat{i} + 3\hat{j} + 3\hat{k})$$

$\therefore$ unit vector along the normal to the given surface is

$$\frac{(-\hat{i} + 3\hat{j} + 3\hat{k})}{\sqrt{1 + 9 + 9}} = \frac{1}{\sqrt{19}} (-\hat{i} + 3\hat{j} + 3\hat{k})$$

Q. If $r^2 = x^2 + y^2 + z^2$, show that $\nabla f(r) = \frac{f'(r)}{r} \vec{r}$.

and hence show that (i) $\nabla r = \hat{r}$ (ii) $\nabla \left(\frac{1}{r}\right) = -\frac{1}{r^3} \vec{r}$
 (iii) $\nabla(r^n) = n r^{n-2} \vec{r}$ (iv) $\nabla \int r^n dr = r^{n-1} \vec{r}$

Solⁿ By defⁿ $\nabla f(r) = \left(i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right) f(r)$

$$= i f'(r) \frac{\partial r}{\partial x} + j f'(r) \frac{\partial r}{\partial y} + k f'(r) \frac{\partial r}{\partial z}$$

$$= f'(r) \left(i \frac{\partial r}{\partial x} + j \frac{\partial r}{\partial y} + k \frac{\partial r}{\partial z} \right)$$

from $r^2 = x^2 + y^2 + z^2$, we have $\frac{\partial r}{\partial x} = \frac{x}{r}$, $\frac{\partial r}{\partial y} = \frac{y}{r}$, $\frac{\partial r}{\partial z} = \frac{z}{r}$

$$\therefore \nabla f(r) = f'(r) \left[i \frac{x}{r} + j \frac{y}{r} + k \frac{z}{r} \right] \text{ or } \boxed{\nabla f(r) = \frac{f'(r)}{r} \vec{r}} \quad \text{--- (1)}$$

(i) Take $f(r) = r$, then from (1), $\nabla r = \frac{1}{r} (1) \vec{r} = \hat{r}$

(ii) Take $f(r) = \frac{1}{r}$, then from (1), $\nabla \left(\frac{1}{r}\right) = \frac{1}{r} \left(-\frac{1}{r^2}\right) \vec{r} = -\frac{1}{r^3} \vec{r}$

(iii) Let us take $f(r) = r^n$, hence from (1) we get

$$\nabla(r^n) = \frac{1}{r} (n r^{n-1}) \vec{r} \text{ or } \boxed{\nabla(r^n) = n r^{n-2} \vec{r}}$$

(iv) we take $f(r) = \int r^n dr$ so that $f'(r) = r^n$

$\therefore$ using (1), here, we get $\nabla \int r^n dr = \frac{1}{r} \cdot r^n \vec{r} = r^{n-1} \vec{r}$